



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2022

SUBJECT : CSE 2154 - OBJECT ORIENTED PROGRAMMING

OBJECT ORIENTED PROGRAMMING [CSE 2154]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A)

```
void fun(int... a)
{
    System.out.println("Number of arguments: " + a.length);
    for (int i : a)
        System.out.print(i + " ");
    System.out.println();
}
```

 (3)

Analyse the code snippet given above and explain the concept behind the given code. Also analyse the following versions of the method signatures and discuss its possible outputs.

```
void fun(int... a, String s)
void fun(int... a, float ... f)
```

- 1B) Demonstrate the use of *continue* and *break* statement with an example program. (3)
- 1C) Define two classes A and B each having one String private data member. Write zero argument and parameterized constructors to initialize the data member and member functions to get, set and display the data member for these classes. In the demo class, write a concat() function which takes the 2 arguments as class A and class B objects and concatenates the data members of these two objects. Write main method that creates instances of the classes A and B, assigns values to each and concatenates the value of data member of these objects using the concat() function. Display the objects before and after concatenation. (4)
- 2A) Create a class Shape with a method void area(). Extend this class Shape to three more classes Rectangle, Triangle and Circle and override the method area() to compute the area of specific shape. Design a Bounded Generic class called BoundGen, which is bounded to Shape class and create a suitable constructor to instantiate object of any sub class of Shape and a print() method which will print area of a specific Shape. Refer the given code Snippet for your reference. (5)
- Sample Code Snippet.
- ```
class Test{
 public static void main(String args[]){
 BoundedGen< Shape> bobj=new BoundedGen<>(new Shape());
 bobj.print();//prints area of Shape
 BoundedGen< Shape> sobj=new BoundedGen<>(new Triangle(2,3));
 sobj.print();//prints the area of Triangle
 BoundedGen< Triangle> tobj=new BoundedGen<>(new Triangle(2,3));
 tobj.print();//prints the area of Triangle
```

```
BoundedGen< Rectangle> robj=new BoundedGen< >(new Rectangle(2,3));
robj.print();//Prints the area of Rectangle
}}
```

- 2B) Illustrate the restrictions while using generics in Java? (3)
- 2C) Illustrate with Non-compliant and compliant code snippets any two rules of Numeric types and Operations with respect to Java CERT. (2)
- 3A) An organization pays its employees on a weekly basis. The employees are of three types: i) Salaried employees are paid a fixed weekly salary regardless of the number of hours worked, ii) hourly employees are paid by the hour and receive overtime pay (i.e., 1.5 times their hourly salary rate) for all hours worked more than 40 hours, iii) commission employees are paid a percentage of their sales. (5)  
Design a class hierarchy with a parent class **Employee** having a private instance variable **Name**, include constructors in each class and two methods, one to print the employee name and the other to compute the salary. Write a test class to show all the functionalities. Use dynamic method dispatch.
- 3B) Create a package testpkg consisting of two classes A and B, class A being public should have three variables, three methods (each variable and method of different access specifier: default, public and protected) and a member interface consisting of two methods. Class B being default should have a single method. Name this program as A.java. (3)  
Write a Java program to compile and execute the package testpkg using fully qualified name.
- 3C) Summarize the restrictions on the members of an interface. Demonstrate "group of constants that will be used by many classes" with an example code. (2)
- 4A) What is Synchronization? Explain different ways of achieving synchronization in Java Multithreading? (2)
- 4B) Design an Interface BookTicket with an instance variable seatsavailable, and a method void reserveTkt(int numberofseats) and CancelTkt(int numberofseats ). Implement the methods in a class AdvBooking. Analyse the above scenario with a minimum of Three Threads, where two threads is trying to reserve a seat and another thread is trying to cancel the reserved tickets. (3)
- 4C) Create a javafx application to prebook a car order in advance as shown in the given Figure. Write the complete program with import statements Analyse the code for proper inputs. (5)

5A) Write a JavaFX application program to display a circle and square using a canvas. Provide a combo box that contains options to draw square and circle shapes. Set the title of the application to Shapes. Upon the press of a button called Change Color, the displayed shapes should change their fill in the order of red, green, blue, yellow and repeat indefinitely. (4)

5B) What is the significance of an abstract class? Illustrate with an example program. (3)

5C) **A.java** and **B.java** are packaged in **pkg1** and **pkg2**, respectively. The code of each files are shown below:

**A.java:**

```
package pkg1;
class A{
private int x;
public int y;
protected int z;
int w;
}
```

**B.java**

```
package pkg2;
class B{
A a = new A();
System.out.println(a.x);
System.out.println(a.y);
System.out.println(a.z);
System.out.println(a.w);
}
```

(3)

When the files are compiled, **B.java** results in errors. Without changing the access modifiers of variables defined in **class A**, rectify the code(s) to prevent any compilation errors. Provide necessary reasons for the change(s) done.

-----End-----